

CS 499 Module Two Journal

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CS 499 Computer Science Capstone

November 8, 2025

Part 1

Code review is the systematic process of examining source code to identify defects, improve readability, and ensure compliance with software design standards before the code is merged into a main repository. It is both a quality assurance and a professional development practice that encourages collaboration and continuous improvement within a software team.

This practice is essential for computer science professionals because it promotes reliable, maintainable, and efficient software development. Code reviews help detect logic errors early in the process, preventing costly fixes later in the project lifecycle. They also strengthen team communication, enhance consistency across large codebases, and reinforce adherence to security and performance standards. Furthermore, they serve as a learning mechanism that allows developers to gain insights from peers and expand their technical skill set.

The best practices identified in this module's resources include keeping reviews focused and manageable, maintaining a respectful and constructive tone, and using automated tools to handle formatting and syntax before human review. Reviews should emphasize logical flow, maintainability, and adherence to project standards over stylistic differences. Code reviews should occur once a developer has completed a specific functional feature but before merging it

into the main branch. Conducting reviews at this point ensures that the code is functional, testable, and ready for peer feedback without introducing unnecessary rework later.

Part 2

For recording my code review, I will use OBS Studio, a versatile open source recording tool that enables me to capture both the source code walkthrough and spoken commentary. OBS provides high quality video and audio, making it ideal for professional presentation and inclusion in my ePortfolio. It will allow me to display my development environment while narrating how my code meets the rubric criteria and professional standards.

To prepare for the recording, I will create a detailed outline for each of my three chosen artifacts. The first artifact, the CS 340 Grazioso Salvare Dashboard, will focus on demonstrating full stack software design principles, database integration, and CRUD operations. My outline for this section will highlight how the MongoDB schema supports data efficiency and how the Python Dash interface provides interactive visualizations for user decision making. The second artifact, my CS 360 Android Inventory Application, will emphasize algorithmic logic, data persistence, and user interface design. The review will showcase how SQLite is used for local storage and how SMS notification functionality was implemented to enhance usability. The third artifact, my CS 370 Reinforcement Learning Agent, will focus on artificial intelligence concepts and algorithm performance. I will review how the Q learning algorithm was optimized for better decision making and how I improved the efficiency of state transitions and reward structures.

In alignment with the rubric, each review section will address three main categories: design and functionality, code quality, and software development process. For design and functionality,

I will demonstrate how each artifact meets user and business requirements through clear and purposeful architecture. In the code quality section, I will analyze clarity, naming conventions, documentation, and logic accuracy. In the software development process section, I will discuss version control usage, testing methodology, and adherence to professional best practices. Outlining these elements in advance will ensure my presentation is organized, comprehensive, and aligned with the CS 499 code review expectations.